



Mate familiarity and social learning in a monogamous lizard

Kirke L. Munch¹ · Daniel W. A. Noble^{1,2} · Erik Wapstra¹ · Geoffrey M. While¹

Received: 24 November 2017 / Accepted: 23 April 2018
© Springer-Verlag GmbH Germany, part of Springer Nature 2018

Abstract

Social learning is thought to be advantageous as it allows an animal to gather information quickly without engaging in costly trial-and-error learning. However, animals should be selective about when and whom they learn from. Familiarity is predicted to positively influence an animal's reliance on social learning; yet, few studies have empirically tested this theory. We used a lizard (*Liopholis whitii*) that forms long-term monogamous pair bonds to examine the effects of partner familiarity on social learning in two novel foraging tasks, an association and a reversal task. We allowed female lizards to observe trained conspecifics that were either familiar (social mate) or unfamiliar execute these tasks and compared these two groups with control females that did not receive social information. Lizards preferentially relied on trial-and-error learning in the association task. In the reversal task, lizards that were demonstrated by familiar partners learnt in fewer trials compared to control lizards and made more correct choices. Our results provide some evidence for context-dependent learning with lizards differentiating between when they utilize social learning, and, to a limited degree, whom they learnt from. Understanding the role of the social context in which learning occurs provides important insights into the benefits of social learning and sociality more generally.

Keywords Cognition · Social learning · Familiarity · Reptiles · *Egernia*

Introduction

The ability of animals to acquire, process and act on information in their environment is fundamental to their fitness (Dayananda and Webb 2017). There are a multitude of ways in which an animal can acquire this information. For

example, an individual can rely on personal information via trial-and-error learning (e.g., asocial learning) or alternatively an individual can rely on information acquired through the observation of and/or interaction with others (e.g., social learning). While social learning was once considered largely restricted to animals living in groups, it is now recognized that a wide range of organisms, not generally thought to be social, utilize social learning strategies (Duffy et al. 2009; Wilkinson et al. 2010a; Noble et al. 2014; Trompf and Brown 2014). Indeed, social learning is predicted to be highly advantageous, allowing individuals to avoid the costs of trial-and-error learning (such as increased effort and risk) (Heyes 1994; Shettleworth 2010; Hoppitt and Laland 2013). Utilizing conspecifics as an information source is thought to be particularly advantageous when the cost of asocial learning is high, especially if there is overlap in resource requirements (such as mate choice decisions and food acquisition) and/or shared predators (Galef and White 1998; Brown and Laland 2003; Lonsdorf and Bonnie 2010).

Theoretical work suggests that, despite its broad benefits, social learning should not be used indiscriminately; instead individuals should adopt strategies that dictate the circumstances under which they copy others, and from whom they

Communicated by Jean-François Le Galliard.

We provide novel evidence that individual characteristics, such as familiarity, mediate the propensity to utilize social information and in more nuanced ways than theory would predict.

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s00442-018-4153-z>) contains supplementary material, which is available to authorized users.

✉ Kirke L. Munch
klmunch@utas.edu.au

- ¹ School of Biological Sciences, University of Tasmania, Private Bag 5, Churchill avenue, Sandy Bay, Hobart, TAS 7005, Australia
- ² School of Biological, Earth, and Environmental Sciences, University of New South Wales, Kensington, NSW 2052, Australia

learn (Laland 2004; Galef and Laland 2005; Heyes 2016; Leris and Reader 2016). Coussi-Korbel and Fragaszy (1995) argue that certain characteristics of the individual conveying information (e.g., the demonstrator), such as age, sex, dominance status, level of success (i.e., mating/foraging), or the relationship to the focal individual, should affect the likelihood of social learning occurring (referred to as ‘directed social learning’). Such context dependent social learning has fundamental implications for the way in which information is transferred through animal populations (Kawai 1965; van de Waal et al. 2010; Aplin et al. 2015; Dubosq et al. 2016). Despite this, relatively few studies have experimentally tested the extent to which individuals discriminate between social information sources (see Nicol and Pope 1994; Swaney et al. 2001; Schwab et al. 2008; Noble et al. 2014; Kar et al. 2017).

Familiarity is one aspect of the social environment that is likely to be particularly important in the context of social learning. Familiar demonstrators are likely to impact focal individual decisions in a number of different ways. First, familiar individuals are more likely to occur in the same temporal and spatial environment; thus, they should convey more accurate information specific to dealing with that particular environment. An individual would therefore benefit more by copying behaviours of familiar demonstrators compared to unfamiliar individuals (Boyd and Richerson 1988; Galef and Giraldeau 2001; Laland 2004). Secondly, familiarity between the focal individual and the demonstrator means that focal individuals will have previous experience regarding the accuracy of the information conveyed (Heyes and Pearce 2015). Indeed, familiarity with a demonstrator results in more effective acquisition of information in a variety of species including fishes (foraging decisions: Lachlan et al. 1998; Swaney et al. 2001; shoaling decision: Griffiths 2003), rodents (predator response: Kavaliers et al. 2005), and birds (foraging decisions: Scheid et al. 2007; Benskin et al. 2002; nest building: Guillette et al. 2016). Despite such studies providing valuable insights into the complexity of social learning and the extent to which social information is conveyed differentially through animal populations, studies that explicitly test the causal role of familiarity in mediating social information transfer are still uncommon.

Social monogamy is a particular form of familiarity whereby males and females pair up over multiple reproductive seasons. One hypothesis for the evolution of stable social monogamy is that it provides benefits in terms of the ability of individuals to produce and raise offspring (‘mate familiarity hypothesis’: Black 1996). This suggests some level of coordination between the social partners, mediated by information transfer (Sanchez-Macouzet et al. 2014; Leu et al. 2015). Here we examined the role that mate familiarity plays in mediating the acquisition and use of social information in a monogamous family living lizard, *Liopholis*

whitii (previously *Egernia whitii*). While social monogamy is relatively rare in reptile systems, species from the Australian *Egernia* group form long-term stable pair bonds (Bull 2000) and vary in their social complexity from largely solitary species to those that live in large stable social groups (reviewed in Chapple 2003). Specifically, *L. whitii* form stable nuclear family groups consisting of long-term socially monogamous pairs and their offspring (Chapple and Keogh 2005; While et al. 2009b). Pairs remain together throughout the year sharing territories and shelter sites and fending off conspecific intruders, whereas offspring disperse within the first year or prior to reaching reproductive maturity (While et al. 2009a). Pair stability between years is high with some pairs remaining together for nine years (While et al. 2009b, GMW unpublished data). As a result of this pair stability, individuals within a population (i.e., the surrounding lizards outside the pair) vary in their degree of familiarity providing us with a unique opportunity to examine the effect of familiarity on social learning in a natural lizard system.

We tested whether familiarity impacts social learning in *L. whitii* using two social learning experiments. We compared the performance of female lizards that had access to trained demonstrators that were either their familiar pair-partner or an unknown male to females that had no demonstration. Based on the prolonged association between pairs in this species, and the potential benefits of relying on social information from closely affiliated individuals, we predicted that individuals with access to social learning would learn in fewer trials in both learning tasks, and individuals with access to social demonstration from familiar partners should learn more readily than those with an unfamiliar demonstrator.

Methods

Study species

Liopholis whitii is a medium sized [up to 100 mm snout-vent length (SVL)], viviparous lizard found throughout a broad altitudinal (0–1600 m) and habitat (coastal heaths, grasslands, and forests) range in south-eastern Australia (Chapple 2003; Wilson and Swan 2003). Males and females are sexually monomorphic and reach reproductive maturity at approximately 3 years and have an overall lifespan of 10–15 years (GMW unpublished data).

Animal capture and husbandry

At the start of November 2015, we caught 124 *L. whitii* (62 males, 62 females) from wild populations on the east coast of Tasmania (42°57'S, 147°88'E). We selectively targeted 20 mating pairs ($n = 40$ individuals) that would remain together

throughout the experiment. Mating pairs were determined based on shared crevice site use, which is a defining feature of *Egernia* mating pairs (Chapple 2003; While et al. 2015) that has been used previously to define social mates (While et al. 2009a, b; Halliwell et al. 2017 for similar approaches). Once captured, sex was determined by eversion of the hemipenes, and lizards were individually marked with non-toxic metallic marker (Artline 990 XF Silver) before being transported in cool, damp cloth bags to the University of Tasmania (~1 h drive). At the University, lizards were weighed (± 1 mg) and measured for SVL and total length (± 0.5 mm). Lizards were then housed in opaque tubs [57 (L) $\times$ 38 (W) $\times$ 32 (H) cm], provided with cat litter as a substrate (~5 thick) and a rock as a basking site and a refuge. Basking lights (25 W) and overhead UV lighting were set to 0800–1800 h night/day cycle to provide thermoregulatory opportunities. Lizards had constant access to water and were fed twice daily with mealworms (*Tenebrio molitor*) as part of the experimental protocol, with the exception of demonstrators in the ‘control’ treatment who were provided with two mealworms every second day (see below). At the completion of the experiment, lizards were released back into the natural populations from which they were captured.

Social learning experiments

We established three treatment groups reflective of the nature of the relationship between the focal individual and their demonstrator that would occur naturally (While et al. 2009b). In group one, social information was provided by the focal individual’s (familiar) partner ($n=20$ mating pairs, hereafter ‘social familiar’). In group two, social information was provided by an unfamiliar individual to the focal individual ($n=22$ pseudo-randomly paired males and females, hereafter ‘social unfamiliar’). Males in this group were rotated every 14 days to maintain a level of unfamiliarity between the male and female pairings. While a previous study has shown that individuals can become “familiar” with one another over relatively short periods of time (Wilkinson et al. 2010b), this familiarity is very different to the level of familiarity that individuals have as a result of the long-term pair bonding observed in this species. Group three was a control group in which the focal individual viewed a random conspecific but received no social information regarding the novel learning tasks [i.e., the control demonstrator did not perform the task ($n=20$ pseudo-randomly paired males and females; males rotated between females every 14 days, hereafter ‘control’)]. To remove potential sex-effects, we used only females as ‘observers’ and males as ‘demonstrators’. Male–female pairs were housed together in the female home cage but were divided with both a fixed transparent Perspex and a removable opaque Perspex. Due to the placement of the fixed transparent barrier, recognition of familiarity and

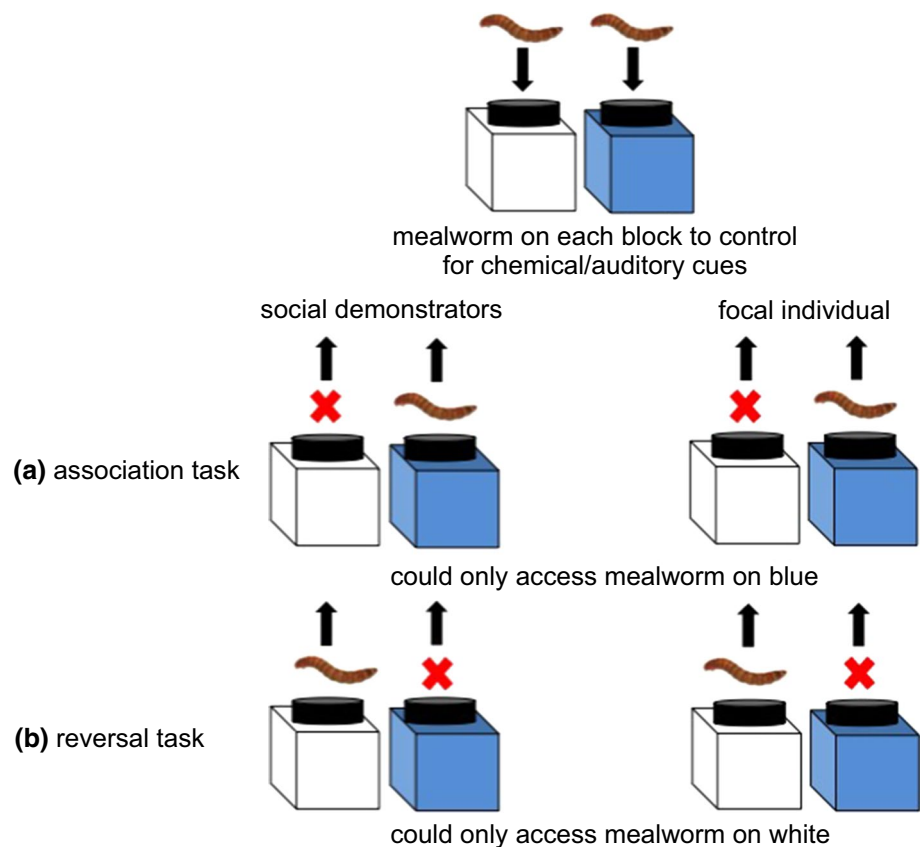
unfamiliarity between the pairs was predominantly visual during the task. However, substrate exchange underneath both the fixed and the removable barriers allowed lizards access to chemical cues about their partner, and we often observed lizards exploring the boundary whilst tongue flicking excessively. All trials were conducted in home cages, and video recorded using a CCTV system (H.264 DVR with Sony 1/2" high resolution colour cameras). We conducted experiments twice-daily, in the morning (0900–1030 h) and afternoon (1300–1430 h) with a minimum interval between trials of 1.5 h. Trials were run daily, except every 14th day when demonstrators from the social unfamiliar and control treatment were rotated.

Our two social learning experiments consisted of an association task with a reversal and were based on learning tasks previously used with lizards (Clark et al. 2013; Noble et al. 2014). Each task consisted of two blocks of wood [7 (L) $\times$ 7 (W) $\times$ 4 (H) cm] being placed in front of the lizard. A petri dish (6 cm diameter) was placed on top of each block using putty (BluetaK®) and was used as a food-well. In addition to dishes being elevated, we wrapped each dish in black, opaque tape to ensure that lizards could not use visual cues during the experiment. We placed a mealworm (~3 cm long) in each of the dishes for both the social demonstrators and the focal lizards to control for chemical and auditory cues, but the reward could only be accessed in one of the dishes. We used fiberglass window screening (601 mm mesh screen) to block access to the mealworm on the opposite block. The block placed on the right side was always blue, and the block placed on the left was always white. The position of the blocks did not change throughout the experiments. This was done deliberately to accelerate learning, as our goal was not to understand whether lizards were using colour or spatial cues to learn. Given sample size constraints, we were not able to counter balance the colour cues (i.e., when a subsample within each treatment and task were given either white or blue as the reward dish). While it is possible that lizards learnt one colour more easily than the other, there is no evidence that this occurs in this species (Munch et al. 2018). Despite this, we checked individual choice data for any biases toward a colour or side prior to analysis and found no obvious biases (see “Statistical analysis”).

Demonstrator training

Before commencing either of the social learning experiments, we trained the demonstrators from the ‘social familiar’ and ‘social unfamiliar’ treatment ($n=42$) to access a food reward from a specific coloured block. For the association task they had to access the food reward from the blue block, whereas for the reversal task, they had to access the reward from the white block (Fig. 1). To solve the task, the lizard had to climb the block and access the food reward

Fig. 1 Tasks presented to social demonstrators and focal (observer) lizards in both the **a** association task and **b** reversal task. (Online version in colour)



hidden in the opaque food dish. The task was designed to be relatively easy compared to those used in other learning studies on lizards (e.g., manipulation of lids; Clark et al. 2013; Noble et al. 2014; Kar et al. 2017) as previous experiments with this species showed that they are unwilling to manipulate objects (KLM unpublished data). This most likely reflects the foraging behaviour of *L. whitii* where they are predominantly ambush predators (i.e., waiting for prey to pass their crevice site) rather than active foragers (GMW pers. obs.). We considered lizards to have learnt the task when they chose the correct block first in 7/8 sequential trials. We continued to give them the task after they achieved this learning criterion in order to assess the robustness of the learning criterion (only for the association task) but capped the number of trials at 75; this was not done for the reversal task due to time restrictions (see supplementary material). Lizards learnt the criterion within 7–75 trials in the association task (mean $\pm$ standard errors (SE): 29.90 ± 3.54 ; $n = 42$) and within 10–161 trials in the reversal task (mean $\pm$ SE: 64.21 ± 7.06 ; $n = 29$). Other studies have previously reported that lizards often need more trials (up to twice as many) to learn a reversal task given the increased complexity of undoing previously associations (e.g., Clark et al. 2013; Kar et al. 2017). We expected that demonstrators that learnt would make some incorrect choices during the experiment so we choose to control for this in our analysis (see “Statistical

analysis”). In theory, however, incorrect choices might help facilitate learning as it shows the observers which options are wrong (Beauchamp and Kacelnik 1991).

Association task

Once we had trained the demonstrator lizards in the association task we then moved onto testing the focal lizard’s ability to learn that specific task. The association task required focal lizards ($n = 62$, $n = 20$ ‘social familiar’, $n = 22$ ‘social unfamiliar’, $n = 20$ ‘control’) to associate a food-reward with the blue block (Fig. 1a). Each task consisted of two phases: 1) a demonstration phase in which the focal individual was allowed to observe the demonstrator completing the task, and 2) a social learning phase in which the focal individual attempted to complete the task. During the demonstration phase, we removed the opaque Perspex (the fixed, transparent Perspex still in place) between the demonstrating and focal lizards to provide an unobstructed view of the demonstrating lizard executing the task (social treatments) or just the conspecific lizard (control). After 30 min of viewing, the trial entered the social learning phase in which we replaced the opaque divider, added the blocks to the focal lizard’s cage and allowed it to attempt the task. We gave the focal lizards a 30-min observation period, as we knew from the demonstrator training that this was roughly the

time it took lizards with previous experience with the task to choose correctly without allowing for multiple choices. The experimental setup was identical between demonstrator and focal lizards except that we capped the number of trials at 75 (based on the demonstrator training). We used a two-step learning criteria to assess whether lizards had learnt the task. First, they had to successfully chose the blue block as their first choice in 7/8 sequential trials. We then ensured that they were 'true' learners by checking that they maintained their learning (i.e., 70% correct choices) for a further 7 trials. Those that did not maintain their learning were not classified as having learnt the task. See supplementary material for robustness of the learning criterion. Focal lizards were allowed to continue with the task even if incorrect choice were initially made; however, these trials did not count towards the learning criterion. Focal lizards reached the learning criterion within 7–58 trials (mean $\pm$ SE: 19.57 ± 1.94 ; $n = 40$). Not all lizards learnt the task ($n = 22$); these were excluded from the subsequent reversal task.

Reversal task

The reversal task required the focal lizards (total $n = 38$; $n = 10$ 'social familiar', $n = 14$ 'social unfamiliar', $n = 14$ 'control') to reverse their previous learning and associate the food reward with the white rather than the blue block (Fig. 1b). Two pairs were excluded due to the escape or death of one of the pair members [$n = 2$; $n = 1$ 'social familiar', $n = 1$ 'unfamiliar']. We first trained the demonstrator lizards in the reversal task and then moved onto testing the focal lizards. As with the previous task, we gave focal lizards a 30-min viewing phase of the demonstrating lizard executing the task (social treatments) or just the demonstrator lizard (control), before allowing them to attempt the task. Focal lizards received more trials (a maximum of $n = 145$ trials) in total for the reversal task as it took some of them longer to reach the learning criterion (range 7–117; mean $\pm$ SE: 51.52 ± 5.94 ; $n = 21$). As with the previous task, we always ensured that they were 'true' learners by checking that they maintained their learning for a further 7 trials (i.e., 70% correct choices). Not all lizards learnt the task ($n = 17$).

Statistical analysis

An observer (KLM) scored all trials for each of the two tasks; the observer was blind to the treatments for the social groups (i.e., the social familiar and unfamiliar treatment) but not to the control treatment, as the 'demonstrator' in this group did not perform the trials in any of the tasks. For each trial, the behaviour scored for both the demonstrator and focal lizard was the lizard's first choice (i.e., whether it chose the correct dish first). A lizard was considered to have made a choice when it placed its front claws on the top

edge of the block. To ensure that there were no unconscious biases, a research assistant re-scored a random selection of 10% of our trials ($n = 365$ trials), while being blind to the hypothesis, original score and the treatment. We assessed score agreement between the two observers using Cohen's Kappa (using the 'psyche' package in R v 3.2.2; Revelle 2017). Cohen's Kappa agreement scores are considered 'excellent' when $k \geq 0.75$ (Kaufman and Rosenthal 2009). Scores agreed 100% of the time ($k = 1$) indicating that our scores were accurate and unbiased.

We analysed our data in three different ways. Lizards that did not reach the learning criterion during a task were excluded from all analyses (association task, total $n = 22$; reversal task, total $n = 17$). First, we compared the number of lizards that learnt (i.e., those achieving the 7/8 correct) in the association and in the reversal tasks using a Fisher's exact test to examine whether our treatments impacted the total number of lizards learning. Second, to compare how quickly lizards learnt in each of the three treatments for both the association and reversal tasks, we compared the mean number of trials taken to reach our learning criteria using a generalised linear model (GLM) with a negative binomial error distribution. Given the logistical constraints in obtaining large samples sizes, which can impact P values, we also calculated Hedge's g to estimate the effect size between the treatment groups (Hedges et al. 1999). We compared the effect sizes for the mean number of trials taken to learn the tasks between: (a) 'social familiar' and 'control' lizards, (b) 'social familiar' and 'social unfamiliar' lizards and (c) 'social unfamiliar' and 'control'. We used Cohen's (1988) benchmarks of small (0.2), medium (0.5) and large (0.8) effects as a guide to interpreting the magnitude of effect size. Third, to test for differences in the rates of learning (i.e., how cognitive performance changed across trials) in our treatments we retained all individual choice data. We then modelled the probability of focal lizards choosing the correct dish first ('1' = yes; '0' = no) as a function of the independent variables, lizard treatment (i.e., 'control', 'social unfamiliar' and 'social familiar'), trial and the interaction between treatment and trial to test for differences in learning rate. We also included a quadratic parameter of trial in the models to test for the possibility that trial number was not necessarily linearly related to the probability of choosing correct. We used generalised linear mixed effects models (GLMMs) with a Bernoulli probability distribution ('logit' link). This approach explicitly allows us to incorporate both between- and within-individual variation in choices, improving power, whilst still allowing us to estimate group-level changes in probability of making correct choices (increases of which provide evidence for learning) in the different treatments (see Kar et al. 2017; Riley et al. 2017 for similar approaches). While mass did not differ significantly between the treatment groups (ANOVA: $F_{2,58} = 1.89$, $P = 0.16$), body

size (SVL) did (ANOVA: $F_{2,59} = 4.04$, $P = 0.02$). While this was only the case for ‘control’ and ‘unfamiliar social’ treatments, we nonetheless included body size to account for the possible effects of SVL on learning (Amiel et al. 2014). To control for non-independence of lizard choices, we estimated a random slope (i.e., trial) and intercept for each lizard (i.e., a random regression model). In all models, we also included an observation-level random effect to account for over-dispersion; this did not impact results, and so, we present models without this variance estimate throughout.

We carried out a number of additional analyses that allowed us to assess the robustness of our results to components of our experimental design. First, we tested whether the choices made by focal lizards in social treatments were impacted by the choices made by the demonstrators. Such effects might be predicted to differ between treatments as lizards in familiar treatments might pay more attention to the choices of the familiar demonstrator compared to lizards in with unfamiliar demonstrators. To test for this possibility, we re-ran the above models while including an interaction parameter between the treatment and the choice made by demonstrators in each trial. This was only run for the social familiar and unfamiliar treatment groups as ‘control’ groups did not have demonstrators for the task. Second, we tested for inherent colour/side bias in individuals. We expected that individuals that choose randomly without any colour/side preferences would have approximately 50:50 correct/incorrect choices within a task before they learnt, and that the number of trials expected to learn would be shorter for the association task, than for the reversal task (Noble et al. 2012; Kar et al. 2017). None of the lizards that learnt in either of the tasks displayed any obvious bias, and we therefore included all lizards that learnt our analyses.

Parameters in all models were estimated using a likelihood framework in R version 3.3.2 (R Core Team 2016). GLMMs were modelled with the *glmer* function in the ‘lme4’ package (Bates et al. 2015), whereas GLMs were run using ‘stats’ package (R Core Team 2016). We z-transformed

all continuous variables (i.e., SVL and trial) prior to analysis by mean centring and dividing by their standard deviation to improve model interpretation (Schielzeth 2010). Nested models were compared using likelihood ratio tests assuming the likelihood ratio followed a Chi square distribution with the degrees of freedom equal to the number of parameters that differed between the models.

Results

Association task

Forty of 62 (67%) lizards were categorized as learners [social familiar, $n = 11$ of 18 (61%); social unfamiliar, $n = 15$ of 22 (68%); control, $n = 14$ of 20 (70%)]. There was no significant difference between treatments in the number of learners (Fisher’s exact test: $P = 0.83$), or between treatment groups in the mean number of trials it took the lizards to learn (Likelihood ratio test (*LRT*): $df = 2$, $\chi^2 = 1.00$, $P = 0.61$; Fig. 2a). Effect sizes were small between the treatment groups (‘social familiar’ vs. ‘control’, Hedge’s $g = 0.18$, $SE = 0.39$; ‘social familiar’ vs. ‘social unfamiliar’, Hedge’s $g = 0.39$, $SE = 0.39$; ‘social unfamiliar’ vs. ‘control’, Hedge’s $g = -0.16$, $SE = 0.36$).

The probability of correctly choosing the reward dish increased nonlinearly across trials (*LRT*: $df = 1$, $\chi^2 = -0.20$, $P < 0.001$; Table 1; Fig. S1a). That is, focal lizards initially increased their probability of choosing correct, only to decrease their number of correct choices in the second half of the task. This suggests that over-training may have occurred in this task (see Carazo et al. 2014). This pattern did not differ between treatments (Trial² × treatment interaction, *LRT*: $df = 2$, $\chi^2 = 0.25$, $P = 0.88$). Demonstrators made a number of incorrect choices during trials. However, controlling for the demonstrator’s choice in analyses of the two social treatments did not affect the probability of the observer choosing correct

Fig. 2 Number of trials (mean $\pm$ SE) until an individual reached the learning criterion (7/8 trials) for **a** the association task and **b** the reversal task. $^{**}P < 0.05$, N.S. not significant for treatment difference. (Online version in colour)

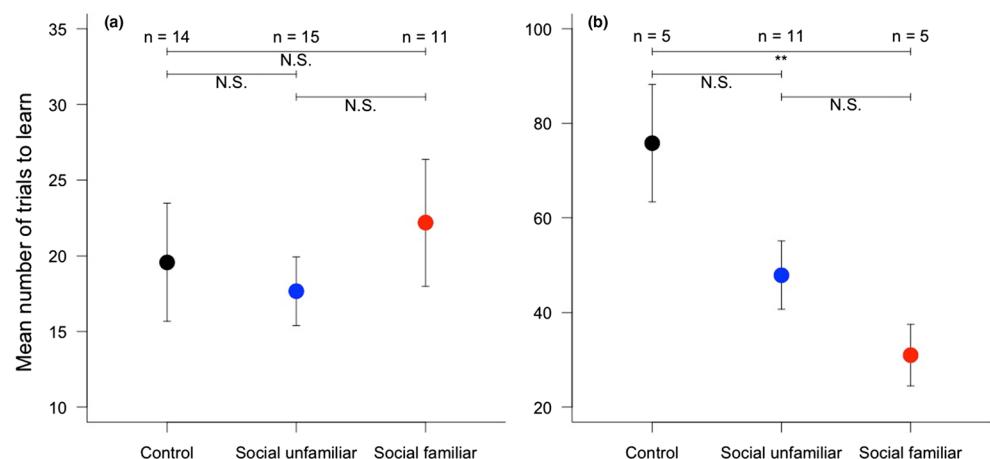


Table 1 Parameter estimates and 95% confidence intervals for generalized linear mixed effects models (GLMMs) examining the effect of familiarity (control, social familiar and social unfamiliar) on social learning in an association and a reversal task with trial modelled as non-linear

	Association task				Reversal task			
	Estimate	Lower	Upper	<i>P</i>	Estimate	Lower	Upper	<i>P</i>
Intercept	1.40	1.06	1.76	< 0.001	0.07	− 0.52	0.54	0.98
Body size (SVL)	0.12	− 0.09	0.32	0.25	− 0.27	− 0.56	0.03	0.07
Trial number	0.18	0.02	0.35	0.03	0.79	0.48	1.17	< 0.001
Trial number ²	− 0.20	− 0.31	− 0.09	< 0.001	0.11	− 0.05	0.28	0.17
Treatment social unfamiliar	− 0.07	− 0.55	0.40	0.78	0.57	− 0.08	1.28	0.08
Treatment social familiar	− 0.27	− 0.77	0.23	0.30	0.86	0.05	1.72	0.03

The models examine the probability for observers of making correct choices (association task: $N_{\text{obs}} = 2463$, $N_{\text{observer lizards}} = 40$, $N_{\text{demonstrator}} = 53$; reversal task: $N_{\text{obs}} = 1193$, $N_{\text{observer lizards}} = 21$, $N_{\text{demonstrator}} = 26$). Statistically significant results are shown in bold

(*LRT*: $\chi^2 = 0.62$, $P = 0.43$) and this did not differ between the two social treatments (demonstrator's choice $\times$ treatment interaction, *LRT*: $\chi^2 = 1.29$, $P = 0.26$; Table S1).

Reversal task

Twenty-one of the 38 (55%) lizards were categorized as learners [social familiar, $n = 5$ of 10 (50%); social unfamiliar, $n = 11$ of 14 (79%); control, $n = 5$ of 14 (36%)]. There was no statistically significant difference between the treatments in the number of learners (Fisher's exact test: $P = 0.08$). However, there was a significant difference between the treatment groups in the mean number of trials it took lizards to learn (*LRT*: $\text{df} = 2$, $\chi^2 = 7.62$, $P = 0.02$; Fig. 2b) with lizards in the 'social familiar' treatment learning the reversal task in significantly fewer trials compared to 'control' lizards (Hedges' $g = -1.82$, $\text{se} = 0.70$). While this effect was weaker when comparing the 'social familiar' to 'social unfamiliar' treatment (Hedges' $g = -0.74$, $\text{SE} = 0.53$) and 'social unfamiliar' to 'control' (Hedges' $g = -1.05$, $\text{SE} = 0.54$), all exhibited moderate to strong effect sizes.

The probability of choosing the correct dish significantly increased across trials in the reversal task (*LRT*: $\chi^2 = 19.01$, $p < 0.001$; Table 1). While the increased probability in choosing the correct dish across trials did not differ significantly between treatment groups (trial $\times$ treatment interaction, *LRT*: $\text{df} = 2$, $\chi^2 = 2.87$, $P = 0.24$), on average, lizards in 'social familiar' treatment made more correct choices than 'control' lizards ($P = 0.04$; Table 1). We found no evidence that the relationship between the probability of choosing correct and trial was nonlinear (Trial², *LRT*: $\chi^2 = 1.89$, $P = 0.17$; Table 1; Fig. S1b). Demonstrator choice did not affect the choice made by the focal (observer) lizard (*LRT*: $\chi^2 = 0.43$, $P = 0.51$) and this did not differ between the two social treatments (demonstrator's choice $\times$ treatment interaction: *LRT*: $\chi^2 = 0.12$, $P = 0.73$; Table S1).

Discussion

We found that *L. whitii* learning from familiar demonstrators use social information to reverse previously learnt associations – learning the reversal task in fewer trials and making more correct choices compared to the control group. This fits with our predictions, namely that individuals with access to social learning would learn in fewer trials, and that lizards with familiar demonstrators would learn more readily. This was not the case for the association task where individuals preferentially relied on trial-and-error learning, suggesting that social learning might not be used indiscriminately. Below we discuss these results in the context of *L. whitii* social system and findings from other species along with the broader implications of our results for our understanding social learning.

We found evidence that lizards expedited learning by making use of social information in the reversal task, but not the association task. Specifically, lizards with access to familiar social information performed better (i.e., learnt quicker and made more correct choices) in the reversal task, learning in nearly half the time compared to the control lizards. The reason why social learning would be utilized to learn a reversal, but not an association task is not entirely clear. One explanation may be that social information is more useful when personal information becomes unreliable as a result of encountering a more complex problem. Indeed, Boyd and Richerson (1988) suggest that individuals will take advantage of relatively cheap information provided by others when personal information becomes costly or difficult to acquire ('costly information hypothesis'). While the costs associated with learning in our foraging tasks are likely minimal, the association task may have been simple enough that trial-and-error learning was a sufficient learning strategy. Our finding of a non-linear relationship between trial and probability of correct choice in this task further supports the idea that the task was simple, as lizards appear to have lost their motivation to engage with the task halfway through the trial process due to over-training (see Carazo

et al. 2014). In contrast, in the reversal task, where the complexity increased, previous foraging knowledge became outdated and it became advantageous to use social information. While we cannot confirm this without additional experimental work (e.g., subjecting different groups of lizards to tasks of varying complexity and measuring their relative use of social vs. asocial learning), previous research has shown that task difficulty influences the particular learning strategy that an individual undertakes (Laland 2004; Kendal et al. 2005). For example, captive *callitrichid* monkeys switch learning strategies when presented with a series of novel puzzle box tasks that varied in difficulty, only relying on social learning to solve the more difficult puzzle boxes (Day et al. 2003). An alternative explanation for our findings may be that lizards were more habituated to the experimental design during the reversal task, allowing them to pay more attention to the demonstrators.

Learning from socially familiar individuals elicited a faster rate of learning compared to lizards without social information, whereas learning from unfamiliar lizards did not elicit as strong of an effect. This provides some support for the suggestion that individuals not only rely on social information but they may do so more from familiar individuals compared to unfamiliar ones. It is possible that by allowing lizard in the ‘social unfamiliar’ treatments to interact with their demonstrators for 14 days we may have dampened the differences between the two social treatment groups, as the lizards would have become more familiar with their demonstrator over time. While additional work is required to confirm the importance of familiarity, given that a limited number of lizards learnt the reversal task overall (dropping power), these results are in line with previous research showing that the identity of the demonstrator can have important implications for the extent to which individuals rely on social vs. asocial information (Nicol and Pope 1994; Swaney et al. 2001; Schwab et al. 2008; Noble et al. 2014). These results also provide some evidence for the ‘mate familiarity hypothesis’ in *L. whittii* (Black 1996), which predicts that social monogamy should select for greater information transfer between long-term partners in order to coordinate reproductive behavior (Sanchez-Macouzet et al. 2014; Leu et al. 2015). Further work, which focuses on the length of the pair bond may provide further insights into the nature of effect of pair familiarity on social learning within this system. Indeed, the actual length of the familiar pair bonds were unknown for our animals and could have varied quite significantly (e.g., pair bond lengths vary from 1 to 9 years in this system; GMW unpublished data) and may mean our result of familiarity-effect on social learning were relatively conservative. Repeating this experiment with lizards of known pair length provides an exciting potential avenue for future research. It would also be interesting to examine whether

there are differences between males and females in their use of social learning and how that related to the identity of the demonstrator, as previous studies have reported sex differences in both brain structure and learning ability in lizards (LaDage et al. 2009; Carazo et al. 2014).

Despite evidence of social information use, the accuracy of the choices made by demonstrators did not appear to influence the performance of the focal lizards. It is generally assumed that individuals should preferentially learn from successful demonstrators (Laland 2004), however, empirical studies have found that this might not always be the case (Beauchamp and Kacelnik 1991; Nicol and Pope 1999; Swaney et al. 2001). Furthermore, any wrong initial choices by the demonstrators may help served to reinforce correct choices later on (Beauchamp and Kacelnik 1991; Templeton 1998).

In summary, we show that individuals that learnt the association task relied on trial-and-error learning but switched learning strategies to make use of social information for the reversal task. Furthermore, we show that in those circumstances, individuals learnt to solve the task in fewer trials and made more correct choices when demonstrated to by familiar demonstrators. While further work is required to confirm these results, our study adds to the growing evidence that individual characteristics, such as familiarity, may mediate the propensity to utilize social information in more nuanced ways than theory would predict (Kendal et al. 2005). Social learning can be affected not only by the social dynamics that exist among individuals, but also by the social setting in which individuals find themselves. We argue that a greater appreciation for the social contexts in which learning is expressed, and explicitly examining not only when but also whom animals learn from, will provide important insight into the fitness benefits of social learning and sociality more generally.

Data availability All data generated and analysed during this study are included in the supplementary files.

Acknowledgements We would like to thank two anonymous reviewers and the handling editor for their constructive feedback on previous versions of the manuscript. We are grateful to Giles Barrington for scoring a subset of our video footage, and we would like to thank the numerous members of the Behavioural and Evolutionary Ecology group that assisted us with lizard collection, husbandry and experimental setup.

Author contribution statement KLM, DWAN, EW and GMW conceived and designed the study. KLM carried out the field and laboratory work. KLM analysed the data with assistance from DWAN and GMW. KLM wrote the paper with assistance from all authors.

Funding This research was supported by the Australian Research Council (DP150102900 to GMW) and a Holsworth Wildlife Endowment Grant (to KLM). KLM was supported by a University of Tasmania

PhD scholarship. GMW (DE150100336), EW (FT110100597), and DWAN (DE150101774) were supported by the Australian Research Council.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All applicable institutional and national guidelines for the care and use of animals were followed. All of the work described here was conducted with the approval of the University of Tasmania's Animal Ethics Committee (A14536).

References

- Amiel JJ, Lindstrom J, Shine R (2014) Egg incubation effects generate positive correlations between size, speed and learning ability in young lizards. *Anim Cognit* 17:337–347. <https://doi.org/10.1007/s10071-013-0665-4>
- Aplin LM, Farine DR, Morand-Ferron J, Cockburn A, Thornton A, Sheldon BC (2015) Experimentally induced innovations lead to persistent culture via conformity in wild birds. *Nature* 518:538–541. <https://doi.org/10.1038/nature13998>
- Bates D, Mächler M, Bolker B, Walker S (2015) Fitting linear mixed-effects models using lme4. *J Stat Softw* 67:1–48. <https://doi.org/10.18637/jss.v067.i01>
- Beauchamp G, Kacelnik A (1991) Effects of the knowledge of partners on learning rates in zebra finches *Taeniopygia guttata*. *Anim Behav* 41:247–253. [https://doi.org/10.1016/S0003-3472\(05\)80476-2](https://doi.org/10.1016/S0003-3472(05)80476-2)
- Benskin CMH, Mann NI, Lachlan RF, Slater PJB (2002) Social learning directs feeding preferences in the zebra finch, *Taeniopygia guttata*. *Anim Behav* 64:823–828. <https://doi.org/10.1006/anbe.2002.2005>
- Black JM (1996) Introduction: pair bonds and partnerships. In: Black JM (ed) Partnerships in birds. The study of monogamy. Oxford University Press, Oxford, pp 3–20
- Boyd R, Richerson PJ (1988) Culture and the evolutionary process. University of Chicago Press, Chicago
- Brown C, Laland KN (2003) Social learning in fishes: a review. *Fish Fish* 4:280–288. <https://doi.org/10.1046/j.1467-2979.2003.00122.x>
- Bull MC (2000) Monogamy in lizards. *Behav Process* 51:7–20. [https://doi.org/10.1016/S0376-6357\(00\)00115-7](https://doi.org/10.1016/S0376-6357(00)00115-7)
- Carazo P, Noble DWA, Chandrasoma D, Whiting MJ (2014) Sex and boldness explain individual differences in spatial learning in a lizard. *Proc R Soc B* 281:20133275. <https://doi.org/10.1098/rspb.2013.3275>
- Chapple DG (2003) Ecology, life-history, and behavior in the Australian Scincid genus *Egernia*, with comments on the evolution of complex sociality in lizards. *Herpetol Monogr* 17:145–180. [https://doi.org/10.1655/0733-1347\(2003\)017\[0145:ELABIT\]2.0.CO;2](https://doi.org/10.1655/0733-1347(2003)017[0145:ELABIT]2.0.CO;2)
- Chapple DG, Keogh JS (2005) Complex mating system and dispersal patterns in a social lizard, *Egernia whitii*. *Mol Ecol* 14:1215–1227. <https://doi.org/10.1111/j.1365-294X.2005.02486.x>
- Clark BF, Amiel JJ, Shine R, Noble DWA, Whiting MJ (2013) Colour discrimination and associative learning in hatchling lizards incubated at 'hot' and 'cold' temperatures. *Behav Ecol Sociobiol* 68:239–247. <https://doi.org/10.1007/s00265-013-1639-x>
- Cohen J (1988) Statistical power analysis for the behavioral sciences. Lawrence Earlbaum Associates, Hillside
- Coussi-Korbel S, Frigaszy DM (1995) On the relation between social dynamics and social learning. *Anim Behav* 50:1441–1453. [https://doi.org/10.1016/0003-3472\(95\)80001-8](https://doi.org/10.1016/0003-3472(95)80001-8)
- Day RL, Coe RL, Kendal JR, Laland KN (2003) Neophilia, innovation and social learning: a study of intergeneric differences in callitrichid monkeys. *Anim Behav* 65:559–571. <https://doi.org/10.1006/anbe.2003.2074>
- Dayananda B, Webb JK (2017) Incubation under climate warming affects learning ability and survival in hatchling lizards. *Biol Lett*. <https://doi.org/10.1098/rsbl.2017.0002>
- Duboscq J, Romano V, MacIntosh A, Sueur C (2016) Social information transmission in animals: lessons from studies of diffusion. *Front Psychol*. <https://doi.org/10.3389/fpsyg.2016.01147>
- Duffy GA, Pike TW, Laland KN (2009) Size-dependent directed social learning in nine-spined sticklebacks. *Anim Behav* 78:371–375. <https://doi.org/10.1016/j.anbehav.2009.05.015>
- Galef BG Jr, Giraldeau L (2001) Social influences on foraging in vertebrates: causal mechanisms and adaptive functions. *Anim Behav* 61:3–15. <https://doi.org/10.1006/anbe.2000.1557>
- Galef BG Jr, Laland K (2005) Social learning in animals: empirical studies and theoretical models. *Bioscience* 55:489–499. [https://doi.org/10.1641/0006-3568\(2005\)055\[0489:SLIAES\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2005)055[0489:SLIAES]2.0.CO;2)
- Galef JBG, White DJ (1998) Mate-choice copying in Japanese quail, *Coturnix japonica*. *Anim Behav* 55:545–552. <https://doi.org/10.1006/anbe.1997.0616>
- Griffiths SW (2003) Learned recognition of conspecifics by fishes. *Fish Fish* 4:256–268. <https://doi.org/10.1046/j.1467-2979.2003.00129.x>
- Guillette LM, Scott ACY, Healy SD (2016) Social learning in nest-building birds: a role for familiarity. *Proc R Soc B* 283:20152685. <https://doi.org/10.1098/rspb.2015.2685>
- Halliwell B, Uller T, Wapstra E, While GM (2017) Resource distribution mediates social and mating behavior in a family living lizard. *Behav Ecol* 28:145–153. <https://doi.org/10.1093/behec/oarw134>
- Hedges LV, Gurevitch J, Curtis PS (1999) The meta-analysis of response ratios in experimental ecology. *Ecology* 80:1150–1156. <https://doi.org/10.2307/177062>
- Heyes CM (1994) Social learning in animals: categories and mechanisms. *Biol Rev Camb Philos Soc* 69:207–231. <https://doi.org/10.1111/j.1469-185X.1994.tb01506.x>
- Heyes CM (2016) Who knows? Metacognitive social learning strategies. *Trends Cogn Sci* 20:204–213. <https://doi.org/10.1016/j.tics.2015.12.007>
- Heyes C, Pearce JM (2015) Not-so-social learning strategies. *Proc Biol Sci*. <https://doi.org/10.1098/rspb.2014.1709>
- Hoppitt W, Laland K (2013) Social learning: on introduction to mechanisms, methods, and models. Princeton University Press, Princeton
- Kar F, Whiting MJ, Noble DWA (2017) Dominance and social information use in a lizard. *Anim Cogn* 20:805–812. <https://doi.org/10.1007/s10071-017-1101-y>
- Kaufman AB, Rosenthal R (2009) Can you believe my eyes? The importance of interobserver reliability statistics in observation of animal behaviour. *Anim Behav* 78:1487–1491. <https://doi.org/10.1016/j.anbehav.2009.09.014>
- Kavaliers M, Colwell DD, Choleris E (2005) Kinship, familiarity and social status modulate social learning about "micropredators" (biting flies) in deer mice. *Behav Ecol Sociobiol* 58:60–71. <https://doi.org/10.1007/s00265-004-0896-0>
- Kawai M (1965) Newly-acquired pre-cultural behavior of the natural troop of Japanese monkeys on Koshima islet. *Primates* 6:1–30. <https://doi.org/10.1007/bf01794457>
- Kendal RL, Coolen I, van Bergen Y, Laland KN (2005) Trade-offs in the adaptive use of social and asocial learning. *Adv Study Behav* 35:333–379. [https://doi.org/10.1016/S0065-3454\(05\)35008-x](https://doi.org/10.1016/S0065-3454(05)35008-x)

- Lachlan RF, Crooks L, Laland KN (1998) Who follows whom? Shoaling preferences and social learning of foraging information in guppies. *Anim Behav* 56:181–190. <https://doi.org/10.1006/anbe.1998.0760>
- LaDage LD, Riggs BJ, Sinervo B, Pravosudiv VV (2009) Dorsal cortex volume in male side-blotched lizards (*Uta stansburiana*) is associated with different space use strategies. *Anim Behav* 78:91–96. <https://doi.org/10.1016/j.anbehav.2009.03.020>
- Laland KN (2004) Social learning strategies. *Learn Behav* 32:4–14. <https://doi.org/10.3758/BF03196002>
- Leris I, Reader SM (2016) Age and early social environment influence guppy social learning propensities. *Anim Behav* 120:11–19. <https://doi.org/10.1016/j.anbehav.2016.07.012>
- Leu ST, Burzacott D, Whiting MJ, Bull CM (2015) Mate familiarity affects pairing behaviour in a long-term monogamous lizard: evidence from detailed bio-bogging and a 31-year field study. *Ethology* 121:760–768. <https://doi.org/10.1111/eth.12390>
- Lonsdorf EV, Bonnie KE (2010) Opportunities and constraints when studying social learning: developmental approaches and social factors. *Learn Behav* 38:195–205. <https://doi.org/10.3758/lb.38.3.195>
- Munch KL, Noble DWA, Botterill-James T, Koolhof IS, Halliwell B, Wapstra E, While GM (2018) Maternal effects on decision-making in a viviparous lizard. *Biol Lett*. <https://doi.org/10.1098/rsbl.2017.0556>
- Nicol CJ, Pope SJ (1994) Social learning in small flocks of laying hens. *Anim Behav* 47:1289–1296. <https://doi.org/10.1016/j.anbehav.2007.06.006>
- Nicol CJ, Pope SJ (1999) The effects of demonstrator social status and prior foraging success on social learning in laying hens. *Anim Behav* 57:163–171. <https://doi.org/10.1006/anbe.1998.0920>
- Noble DWA, Carazo P, Whiting MJ (2012) Learning outdoors: male lizards show flexible spatial learning under semi-natural conditions. *Biol Lett* 8:946–948. <https://doi.org/10.1098/rsbl.2012.0813>
- Noble DW, Byrne RW, Whiting MJ (2014) Age-dependent social learning in a lizard. *Biol Lett* 10:20140430. <https://doi.org/10.1098/rsbl.2014.0430>
- R Core Team (2016) R: A language and environment for statistical computing. Vienna, Austria. R Foundation for Statistical Computing. See <http://R-project.org>
- Revelle W (2017) “psych”: procedures for personality and psychological research. See <https://CRAN.R-project.org/package=psych>, version 1.7.8
- Riley JL, Noble DWA, Byrne RW, Whiting MJ (2017) Does social environment influence learning ability in a family-living lizard? *Anim Cogn* 20:449–458. <https://doi.org/10.1007/s10071-016-1068-0>
- Sanchez-Macouzet O, Rodriguez C, Drummond H (2014) Better stay together: pair bond duration increases individual fitness independent of age-related variation. *Proc R Soc B* 281:7. <https://doi.org/10.1098/rspb.2013.2843>
- Scheid C, Range F, Bugnyar T (2007) When, what, and whom to watch? Quantifying attention in ravens (*Corvus corax*) and jackdaws (*Corvus monedula*). *J Comp Psychol* 121:380–386. <https://doi.org/10.1037/0735-7036.121.4.380>
- Schielzeth H (2010) Simple means to improve the interpretability of regression coefficients. *Methods Ecol Evol* 1:103–113. <https://doi.org/10.1111/j.2041-210X.2010.00012.x>
- Schwab C, Bugnyar T, Kotrschal K (2008) Preferential learning from non-affiliated individuals in jackdaws (*Corvus monedula*). *Behav Process* 79:148–155. <https://doi.org/10.1016/j.beproc.2008.07.002>
- Shettleworth SJ (2010) Cognition, evolution, and behavior, 2nd edn. Oxford University Press, New York
- Swaney W, Kendal J, Capon H, Brown C, Laland KN (2001) Familiarity facilitates social learning of foraging behaviour in the guppy. *Anim Behav* 62:591–598. <https://doi.org/10.1006/anbe.2001.1788>
- Templeton JJ (1998) Learning from others’ mistakes: a paradox revisited. *Anim Behav* 55:79–85. <https://doi.org/10.1006/anbe.1997.0587>
- Trompf L, Brown C (2014) Personality affects learning and trade-offs between private and social information in guppies, *Poecilia reticulata*. *Anim Behav* 88:99–106. <https://doi.org/10.1016/j.anbehav.2013.11.022>
- van de Waal E, Renevey N, Favre CM, Bshary R (2010) Selective attention to philopatric models causes directed social learning in wild vervet monkeys. *Proc R Soc B* 277:2105–2111. <https://doi.org/10.1098/rspb.2009.2260>
- While GM, Uller T, Wapstra E (2009a) Family conflict and the evolution of sociality in reptiles. *Behav Ecol* 20:245–250. <https://doi.org/10.1093/beheco/arp015>
- While GM, Uller T, Wapstra E (2009b) Within-population variation in social strategies characterize the social and mating system of an Australian lizard, *Egernia whitii*. *Austral Ecol* 34:938–949. <https://doi.org/10.1111/j.1442-9993.2009.02002.x>
- While GM, Chapple DG, Gardner MG, Uller T, Whiting MJ (2015) *Egernia* lizards. *Curr Biol* 25:R593–R595. <https://doi.org/10.1016/j.cub.2015.02.070>
- Wilkinson A, Kuenstner K, Mueller J, Huber L (2010a) Social learning in a non-social reptile (*Geochelone carbonaria*). *Biol Lett* 6:614–616. <https://doi.org/10.1098/rsbl.2010.0092>
- Wilkinson A, Specht HL, Huber L (2010b) Pigeons can discriminate group mates from strangers using the concept of familiarity. *Anim Behav* 80:109–115. <https://doi.org/10.1016/j.anbehav.2010.04.006>
- Wilson S, Swan G (2003) A complete guide to reptiles of Australia. Reed New Holland, Sydney